

Submission Worksheet

Submission Data

Course: IT202-008-S2026

Assignment: IT202 PHP Challenge Scenarios

Student: Stephane E. (se283)

Status: Submitted | **Worksheet Progress:** 100+%

Potential Grade: 11.00/10.00 (110.00%)

Received Grade: 0.00/10.00 (0.00%)

Started: 3/2/2026 8:59:27 PM

Updated: 3/2/2026 11:09:23 PM

Grading Link: <https://learn.ethereallab.app/assignment/v3/IT202-008-S2026/it202-php-challenge-scenarios/grading/se283>

View Link: <https://learn.ethereallab.app/assignment/v3/IT202-008-S2026/it202-php-challenge-scenarios/view/se283>

Instructions

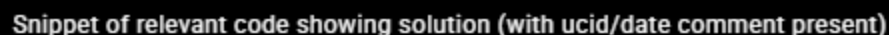
- Overview Link: <https://youtu.be/iwA6vyRPhoM> (older reference, but should be accurate for now)
- 1. Ensure you read all instructions and objectives before starting. It's recommended to solve all the requirements first, then fill out the worksheet.
- 2. Create a new branch from qa called M2-Homework
 - 1. `git checkout qa` (ensure proper starting branch)
 - 2. `git pull origin qa` (ensure history is up to date)
 - 3. `git checkout -b M2-Homework` (create and switch to branch)
- 3. Copy the template code from here: [GitHub Repository - M2 Homework](#)
 - It includes Scenarios 1-4 and `base.php`. Put all into an M2 folder or similar inside your `public_html`
 - Immediately record to history
 - `git add public_html`
 - `git commit -m "adding M2 HW baseline files"`
 - `git push origin M2-Homework`
 - Create a Pull Request from M2-Homework to qa and keep it open
- 4. Fill out the below worksheet
 - Each Scenario requires the following as you work
 - Ensure there's a comment with your UCID, date, and brief summary of how the problem was solved
 - Initial outline/plan of how you'll solve it via comments (add/commit after this stage)
 - Code solution (add/commit periodically as needed)
 - If you chose to tackle this problem by problem and fill out the worksheet as you go, you'll likely need to keep merging pull requests into qa
- 5. Once finished, click "Submit and Export"
- 6. Locally add the generated PDF to a folder of your choosing inside your repository folder (homework folder is fine) and move it to Github
 - 1. `git add .`
 - 2. `git commit -m "adding PDF"`

- Section #1: (2 pts.) Scenario 1 - Odds

≡ Task #1 (2 pts.) - Edit the `printOdds` function to output odd values of the array

Part 1:

1. Snippet of relevant code showing solution (with ucid/date comment present)
2. Full output of executing the program (visit the proper file on Render.com qa after a manual deploy)



Array 1: Original Array: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9 Output Array: 1, 3, 5, 7, 9	Array 2: Original Array: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9 Output Array: 0, 2, 4, 6, 8
Array 3: Original Array: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9 Output Array: 1, 3, 5, 7, 9	Array 4: Original Array: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9 Output Array: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9

Completed Scenario 1 for [se283] [2026-03-03 03:33:04]

Full output of executing the program (visit the proper file on Render.com qa after a manual deploy)

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Part 2:

Progress: 100%

Details:

- Direct link to the file in the homework related branch from Github (should end in .php, , zero credit if it does not)
- Direct link to the file on Render.com Prod (Grab the qa url you're testing, paste it, change "-qa" to "-prod"), zero credit if

URL #1

https://github.com/se283/se283-it202-008-2026-03-03-03-33-04-Homework/public_html/M2/scenario1.php



URL

https://github.com/se283/se283-it202-008-2026-03-03-03-33-04-Homework/public_html/M2/scenario1.php



URL #2

<https://se283-it202-008-prod.onrender.com/M2/scenario1.php>



URL

<https://se283-it202-008-prod.onrender.com/M2/scenario1.php>



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Part 3:

Progress: 100%

Details:

Briefly explain how your solution works. Describe the logic, steps, or decisions your code makes to reach the result. Do not restate the problem or list requirements.

Your Response:

The code works by first creating an empty list to hold the odd numbers. It then uses a loop to look at every number in the original array one by one. For each number, it uses the modulo operator (%) to check if there is a remainder when divided by 2; if the remainder is not zero, the number is odd and gets added to the new list. Finally, the code uses the implode function to turn that list into a single string, putting a comma and a space between the numbers while automatically making sure there are no extra commas at the very beginning or end.

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Progress: 100%

Progress: 100%

Progress: 100%



Full output of executing the program (visit the proper file on Render.com qa after a manual deploy)

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Part 2:

Progress: 100%

Details:

- Direct link to the file in the homework related branch from Github (should end in .php, , zero credit if it does not)
- Direct link to the file on Render.com Prod (Grab the qa url you're testing, paste it, change "-qa" to "-prod"), zero credit if , zero credit if it's not the direct link to the file or doesn't load

URL #1

[https://github.com/se283/se283-it202-008-2025/
Homework/public_html/M2/scenario2.php](https://github.com/se283/se283-it202-008-2025/Homework/public_html/M2/scenario2.php)



[https://github.com/se283/se283-it202-008-2025/
Homework/public_html/M2/scenario2.php](https://github.com/se283/se283-it202-008-2025/Homework/public_html/M2/scenario2.php)



URL #2

<https://se283-it202-008-prod.onrender.com/M2/scenario2.php>



<https://se283-it202-008-prod.onrender.com/M2/scenario2.php>



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Part 3:

Progress: 100%

Details:

Briefly explain how your solution works. Describe the logic, steps, or decisions your code makes to reach the result. Do not restate the problem or list requirements. Additionally, did you notice anything about the math of some of the data? What do you think happened and what's the reason for it?

Your Response:

The code calculates a total by starting a counter at 0 and using a for loop to visit each number in the list, adding each to the sum. After the loop finishes, it uses a tool called number_format() to turn that sum into a "modified" version that always shows exactly two decimal places, making it look like a dollar amount. Some math results may not look right in a conventional way, mainly when adding tiny numbers to huge ones. The reason behind this is that computers have a limit on how many digits they can remember at once, and therefore will essentially forget the tiny number because it doesn't have enough space to store a big value and a small value in the same calculation.

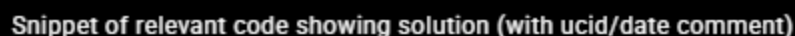
Progress: 100%

Progress: 100%

- Only make edits where noted via provided comments
- Challenge 1: Make each value positive
- Challenge 2: Convert the values back to their original data type and assign it to the proper slot of the output array
- Step 1: sketch out plan using comments (include ucid and date)
- Step 2: Add/commit your outline of comments (required for full credit)
- Step 3: Add code to solve the problem (add/commit as needed)

Progress: 100%

1. Snippet of relevant code showing solution (with ucid/date comment)
2. Full output of executing the program (visit the proper file on Render.com qa after a manual deploy)




```
1.1 (5), 2.2 (5), 3.3 (5), (5), 4.4 (5), 5.5 (5), 6.6 (5), 7.7 (5), 8.8 (5)
123 (5), 444 (5), 555 (5), 666 (5), 777 (5), 888 (5), 999 (5)
Output:
1.1 (5), 2.2 (5), 3.3 (5), 4.4 (5), 5.5 (5), 6.6 (5), 7.7 (5), 8.8 (5)
123 (5), 444 (5), 555 (5), 666 (5), 777 (5), 888 (5), 999 (5)
Completed Scenario 3: For [se283] [2026-08-08 08:08:00]
```

Full output of executing the program (visit the proper file on Render.com qa after a manual deploy)

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Part 2:

Progress: 100%

Details:

- Direct link to the file in the homework related branch from Github (should end in .php, , zero credit if it does not)
- Direct link to the file on Render.com Prod (Grab the qa url you're testing, paste it, change "-qa" to "-prod"), zero credit if

URL #1 https://github.com/se283/se283-it202-008b2026/Homework/public_html/M2/scenario3.php		URL https://github.com/se283/se283-it202-008b2026/Homework/public_html/M2/scenario3.php	
URL #2 https://se283-it202-008-prod.onrender.com/M2/scenario3.php		URL https://se283-it202-008-prod.onrender.com/M2/scenario3.php	

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Part 3:

Progress: 100%

Details:

Briefly explain how your solution works. Describe the logic, steps, or decisions your code makes to reach the result. Do not restate the problem or list requirements.

Your Response:

The code goes through an array by iterating through each element using a for loop. Inside the loop, it first calculates the absolute value of the current element to check if it is positive. It then identifies the element's original data type (such as integer, double, or string) using the `gettype()` function. Finally, the code uses a series of conditional statements to cast that positive value back into its original type before saving it into a specific slot in the output array.

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Section #4: (2 pts.) Scenario 4 - Strings

☰ Task #1 (2 pts.) - Edit the `transformText` function to solve the challenges

Progress: 100%

Details:

- Only make edits where noted via provided comments
- Challenges will be applied collectively
- Challenge 1: Remove non-alphanumeric characters except spaces
- Challenge 2: Convert text to Title Case
- Challenge 3: Remove leading/trailing spaces and remove duplicate spaces between words
- Result 1-3: Assign final phrase to `placeholderForModifiedPhrase`
- Step 1: sketch out plan using comments (include uuid and date)
- Step 2: Add/commit your outline of comments (required for full credit)
- Step 3: Add code to solve the problem (add/commit as needed)

Part 1:

Progress: 100%

Details:

Two screenshots are expected

1. Snippet of relevant code showing solution (with uuid/date comment)
2. Full output of executing the program (visit the proper file on Render.com qa after a manual deploy)

Snippet of relevant code showing solution (with ucid/date comment)



Full output of executing the program (visit the proper file on Render.com qa after a manual deploy)



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Part 2:

Progress: 100%

Details:

- Direct link to the file in the homework related branch from Github (should end in .php, , zero credit if it does not)
- Direct link to the file on Render.com Prod (visit the proper file on Render.com qa after a manual deploy)

URL #1

https://github.com/se283/se283-it202-008b2025/Homework/public_html/M2/scenario4.php



URL

<https://github.com/se283/se283-i>



URL #2

<https://se283-it202-008-prod.onrender.com/M2/scenario4.php>



URL

<https://se283-it202-008-prod.onre>



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Part 3:

Progress: 100%

Details:

Briefly explain how your solution works. Describe the logic, steps, or decisions your code makes to reach the result. Do not restate the problem or list requirements.

Your Response:

To reach the final result, the code first uses a regular expression (Regex) to strip out symbols and special characters while specifically preserving letters, numbers, and spaces. Next, it converts the entire string to lowercase to flatten any irregular capitalization before applying the ucwords() function, which ensures every word starts with a capital letter. Finally, the code wraps up by trimming whitespace from the ends and using another regular expression to find any internal double spaces, collapsing them into single spaces to ensure the phrase is clean and professionally formatted.



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Task #2 (+ 1 pt.) - Edit the `transformText` function to solve the extra credit challenge (challenge 4, Extra Credit)

Progress: 100%

Details:

- Only make edits where noted via provided comments
- Challenge 4: Extract up to middle 3 characters (beginning starts at middle of phrase, exclude the first and last character for shorter phrases)
- Assign result to 'placeholderForMiddleCharacters'
- If not enough characters in a word, instead assign "Not enough characters" to placeholderForMiddleCharacters
- Step 1: sketch out plan using comments (includeucid and date)
- Step 2: Add/commit your outline of comments (required for full credit)
- Step 3: Add code to solve the problem (add/commit as needed)
- Note: There isn't a URL entry for this since it's part of Scenario 4

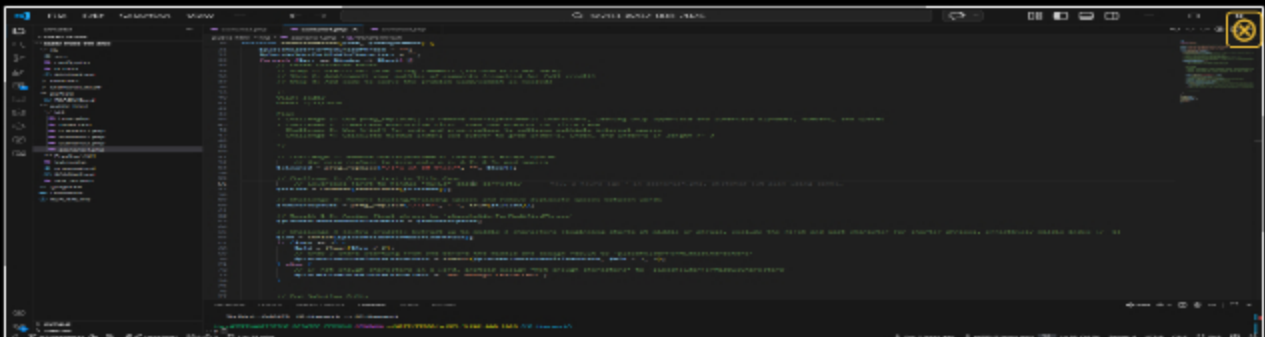
Part 1:

Progress: 100%

Details:

Two screenshots are expected

1. Snippet of relevant code showing solution (withucid/date comment)
2. Full output of executing the program (visit the proper file on Render.com qa after a manual deploy)



```
function extractMiddleCharacters(word) {
  // Extract up to middle 3 characters (beginning starts at middle of phrase,
  // exclude the first and last character for shorter phrases)
  // Assign result to 'placeholderForMiddleCharacters'
  // If not enough characters in a word, instead assign "Not enough characters" to
  // placeholderForMiddleCharacters
  // Step 1: sketch out plan using comments (includeucid and date)
  // Step 2: Add/commit your outline of comments (required for full credit)
  // Step 3: Add code to solve the problem (add/commit as needed)
  // Note: There isn't a URL entry for this since it's part of Scenario 4

  // Your Solution Logic
  let placeholderForMiddleCharacters = "Not enough characters";

  if (word.length > 3) {
    // Extract up to middle 3 characters (beginning starts at middle of phrase,
    // exclude the first and last character for shorter phrases)
    placeholderForMiddleCharacters = word.substring(Math.floor(word.length / 2) - 1, Math.floor(word.length / 2) + 3);
  }

  return placeholderForMiddleCharacters;
}
```

Snippet of relevant code showing solution (withucid/date comment)



```
function extractMiddleCharacters(word) {
  // Extract up to middle 3 characters (beginning starts at middle of phrase,
  // exclude the first and last character for shorter phrases)
  // Assign result to 'placeholderForMiddleCharacters'
  // If not enough characters in a word, instead assign "Not enough characters" to
  // placeholderForMiddleCharacters
  // Step 1: sketch out plan using comments (includeucid and date)
  // Step 2: Add/commit your outline of comments (required for full credit)
  // Step 3: Add code to solve the problem (add/commit as needed)
  // Note: There isn't a URL entry for this since it's part of Scenario 4

  // Your Solution Logic
  let placeholderForMiddleCharacters = "Not enough characters";

  if (word.length > 3) {
    // Extract up to middle 3 characters (beginning starts at middle of phrase,
    // exclude the first and last character for shorter phrases)
    placeholderForMiddleCharacters = word.substring(Math.floor(word.length / 2) - 1, Math.floor(word.length / 2) + 3);
  }

  return placeholderForMiddleCharacters;
}
```

Full output of executing the program (visit the proper file on Render.com qa after a manual deploy)

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Part 2:

Details:

Briefly explain how your solution works. Describe the logic, steps, or decisions your code makes to reach the result. Do not restate the problem or list requirements.

Your Response:

To find the middle three characters, the code first checks if the text is at least three letters long. It then divides the total length by two and rounds down to find the exact center spot of the phrase. By starting one step before that center and pulling out a three-letter piece, the code pulls out a small section of text. If the phrase is too short to have a middle, it simply labels it as not having enough characters to avoid a computer error.



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Section #5: (2 pts.) Misc

Progress: 100%

≡ Task #1 (0.67 pts.) - Github Details

Progress: 100%

Part 1:

Progress: 100%

Details:

From the Commits tab of the Pull Request, screenshot the commit history Following minimum should be present: addition of baseline files, comments planning thoughts, solutions



From the Commits tab of the Pull Request, screenshot the commit history Following minimum should be present: addition of baselin



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Part 2:

Progress: 100%

Details:

Include the link to the Pull Request (should end in /pull/#, zero credit if it does not)

URL #1

<https://github.com/se283/se283-it202-004-2026/>



UH

<https://github.com/se283/se283-i>



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Task #2 (0.67 pts.) - WakaTime - Activity

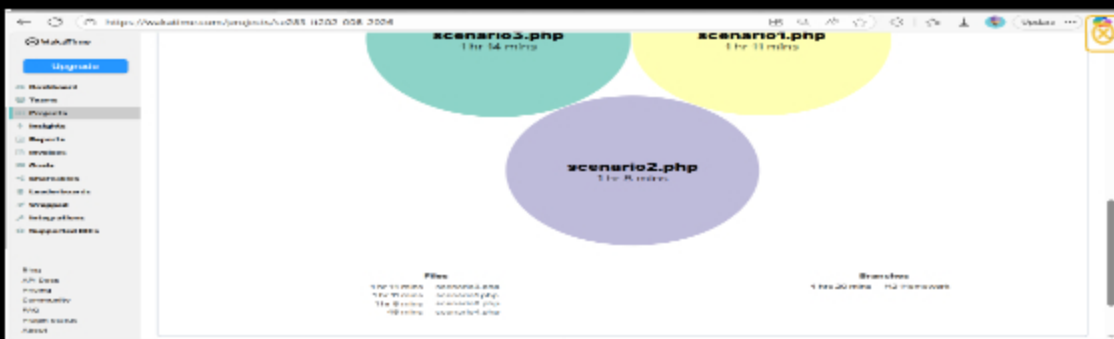
Progress: 100%

Details:

- Visit the [WakaTime.com](https://wakatime.com) Dashboard
- Click Projects and find your repository
- Capture the overall time at the top that includes the repository name
- Capture the individual time at the bottom that includes the file time
- Note: The duration isn't relevant for the grade and the visual graphs aren't necessary



Capture the overall time at the top that includes the repository name



Capture the individual time at the bottom that includes the file time



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≡ Task #3 (0.67 pts.) - Reflection

Progress: 100%

⇒ Task #1 (0.33 pts.) - What did you learn?

Progress: 100%

Details:

Briefly answer the question with at least a few reasonable sentences

Your Response:

One aspect of PHP that I learned is how the implode() function works in the PHP challenge scenario 1. Another aspect I learned is using the gettype() function to get the data type of a variable in the PHP challenge scenario 3. Finally, I also learned about incorporating Regex in PHP through the preg_match() function in PHP challenge scenario 4.



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⇒ Task #2 (0.33 pts.) - What was the easiest part of the assignment?

Progress: 100%

Details:

Briefly answer the question with at least a few reasonable sentences

Your Response:

The easiest parts of the assignment are the PHP challenge scenarios 1 and 2. I find the tasks in those first two PHP challenges relatively simpler to solve than in the others, mostly because they use basic for loops. In Scenario 1, I used a loop to check each number and find the odd ones. In Scenario 2, I used a loop to add all the numbers together and then formatted the final sum to show two decimal places.



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⇒ Task #3 (0.33 pts.) - What was the hardest part of the assignment?

Progress: 100%

Details:

Briefly answer the question with at least a few reasonable sentences

Your Response:

The hardest part of the assignment was the PHP challenge scenario 4. In PHP challenge scenario 4, handling the "Middle 3" extraction is difficult because it requires precise index math for both even and odd string lengths. You have to calculate a middle point, subtract one to find the starting position, and ensure you don't accidentally grab characters outside the string's range. Additionally, combining multiple PHP functions like preg_replace and ucwords in the correct order is tricky; if you don't lowercase the text before applying Title Case or if your regex accidentally deletes spaces along with special characters, the final phrase won't look right.



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